

LCD DISPLAY RECOGNITION SYSTEM

System Design Specification v1.1

Detection Architecture, Annotation Standard,
Text Ontology, Symbol Catalog and Dataset Specification

Document control	Value
Status	Final Phase 1 baseline
Prepared for	Management, ML Engineering and Annotation Teams
Date	14 September 2026
Text model baseline	YOLO multi-class object detector
Symbol capability	Existing symbol-recognition subsystem
Master annotations	JSON with pixel-coordinate bounding boxes
AUTHORITATIVE PURPOSE	
Defines the text-plus-symbol problem, system boundary, integration architecture, ontology, annotation rules, JSON contract, dataset lifecycle, metrics and version controls.	

02 Executive Summary

System intent and key engineering decisions

The system will extract all meaningful LCD content from an upright cropped electricity-meter display. Text is detected through a closed-vocabulary object detector. Approved symbols are supplied by the completed symbol-recognition workstream. Both outputs are merged into one coordinate-consistent JSON record.

PHASE 1 GOAL

Return visible text and approved symbols with class labels and bounding boxes. Do not translate the display into business-level parameter meaning.

Item	Decision
Input	Complete crop of active LCD glass.
Text method	Detection-first: digit-wise values, separate punctuation, whole-token labels and units.
Symbols	Existing subsystem integrated into final output.
Output	Unified JSON with text and symbol detections.
Reconstruction	Human-readable text generated from spatial ordering.
Generalization	No fixed meter coordinates or per-layout templates.

- Master JSON is the source of truth; YOLO TXT and COCO JSON are exports.
- Decimal and colon recall are release-critical.
- Symbol descriptions use stable canonical names.

03 Detailed Problem Definition

Scope, assumptions and required behavior

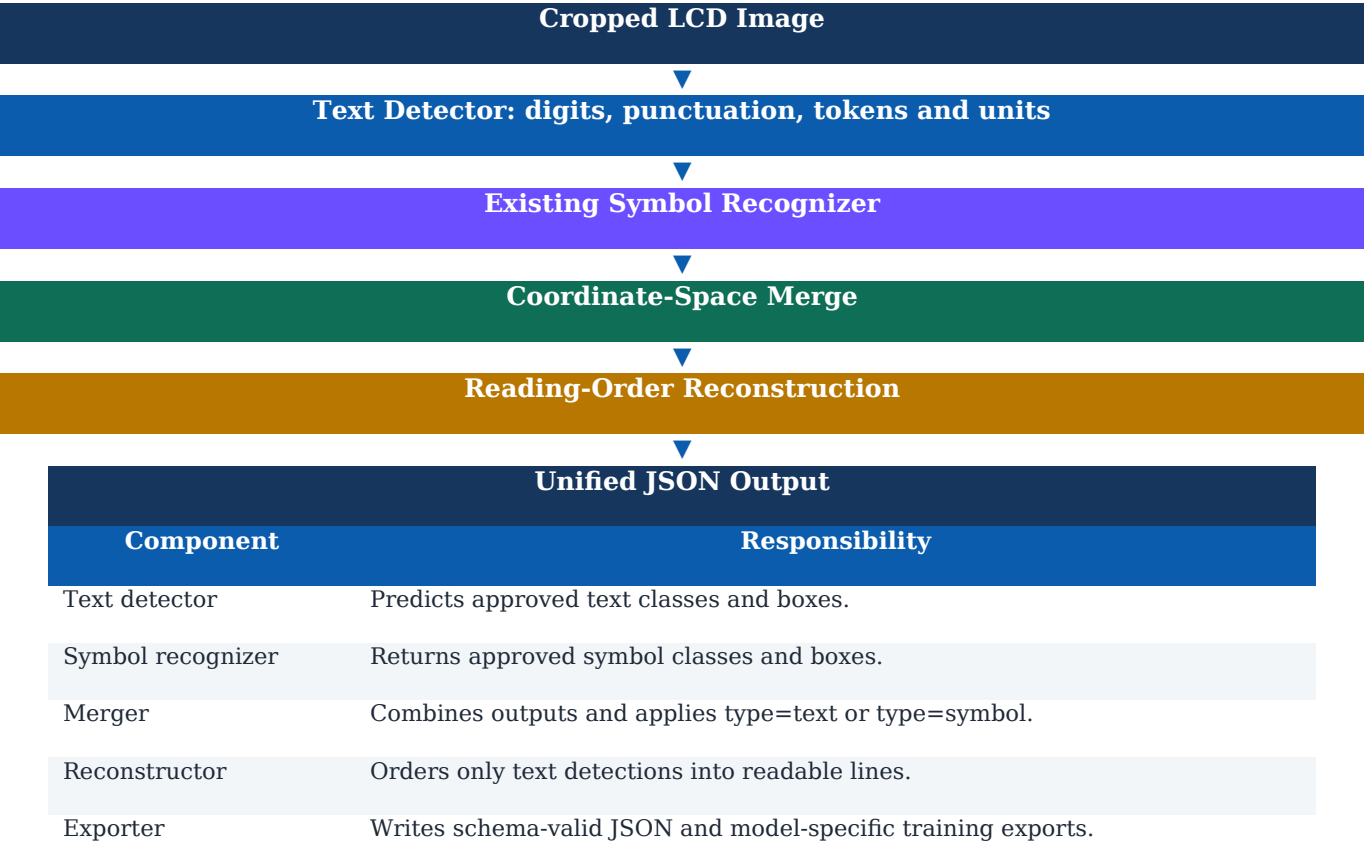
Field	Requirement
Crop	Active LCD glass plus a small consistent margin.
Orientation	Upright and straight.
Technology	7-segment and 14-segment LCD.
Layout	Variable across utilities and meter families.
Quality	No camera blur or glare; display-native unevenness is in scope.
Localization	LCD already localized.
Content	Digits, punctuation, fixed text, units and approved symbols.

- Detect every visible approved text element.
- Preserve repeated instances.
- Return tight pixel bounding boxes.
- Merge existing symbol detections without changing coordinate space.
- Produce deterministic JSON and reconstructed text.

OUT OF SCOPE
Business interpretation, parameter-value linkage, raw meter localization, perspective correction, camera-condition robustness and training a new symbol model.

04 Integrated Architecture

Text detection and existing symbol recognition



05 Detection-First Rationale

Why OCR is not the primary Phase 1 method

Criterion	Value-level OCR	Detection-first
Bounding boxes	Often word/line level.	Native per element.
Decimal	Can be dropped or mapped to a letter.	Independent decimal class.
Colon	Can be confused with decimal.	Independent colon class.
Fixed labels	Decoded character by character.	Stable whole-token class.
Debugging	Sequence errors are less local.	Every error has class and location.
Annotation cost	Lower.	Higher one-time cost.
Requirement fit	Partial.	Direct.

FALLBACK PATH

If the pilot detector cannot achieve acceptable punctuation recall, retain the same JSON schema and evaluate a hybrid value recognizer. The dataset and symbol subsystem remain reusable.

Recommended text baseline: a YOLO small model for the pilot, followed by a medium model if capacity or small-object recall is insufficient. Input size must be selected after measuring the smallest punctuation boxes.

06 Unified Ontology

Text families and symbol family

Family	Classes or source
Digits	0 1 2 3 4 5 6 7 8 9
Punctuation	. :
Fixed text tokens	A, Alr, AutO, AVG, B, B1, BAL, BA, CUM, C_OPEN, EP, F, FG, FOR, IMD, IP, L1, L2, lr, MD, NEU, P, PF, POB, PO mode n, Pr mode F, Pr mode n, PuSh, R, S, T1, T2, T3, T4, Y
Units	A, V, Hz, kW, kWh, kVA, kVAh
Symbols	Canonical approved classes from the completed symbol subsystem.

- Class IDs are immutable after release.
- Names are case-sensitive.
- New classes are appended, never inserted.
- A class requires visible examples and an approved description.
- Text and symbol classes remain distinguishable through the detection type field.

07 Text Annotation Standards

Rules for detector ground truth

ID	Rule	Requirement
A1	Digit-wise	Every digit receives its own box.
A2	Decimal	Visible decimal receives class "." and a tight box.
A3	Colon	One box covers both colon dots and receives class ":".
A4	Longest token	Use PF, L1 and kWh as whole approved classes.
A5	Repeated objects	Each occurrence is annotated.
A6	Tightness	Minimal consistent padding around lit glyphs.
A7	Ambiguity	Do not guess unreadable content.
A8	Unknown class	Flag for ontology review; never invent.
A9	Symbols	Do not duplicate annotations already owned by the symbol subsystem.
A10	Coordinates	Use [x1,y1,x2,y2] in original image pixels.
GOLDEN RULE Annotate what is visible. Do not normalize, repair, translate or interpret the LCD.		

08 Good vs Bad Annotation

Canonical examples for consistent labels

Visible	Good	Bad	Reason
PF 0.998	PF 0 . 9 9 8	PF 0.998	Value is not one object.
12:45:31	1 2 : 4 5 : 3 1	12 : 45 : 31	Digits are separate.
000030 kWh	0 0 0 0 3 0 kWh	000030 k W h	kWh is a whole token.
L1 L2	L1 L2	L 1 L 2	Use approved tokens.
C_OPEN	C_OPEN	C _ OPEN	Use closed class.
0206.32	0 2 0 6 . 3 2	0 2 0 6 3 2	Decimal is mandatory.
COLON RULE A colon is one object containing both dots. Never annotate its two dots as decimal points.			

09 Symbol Description Catalog

Canonical classes for the completed symbol workstream

Canonical symbol	Description
Earth_Load_Tamper	Earth-load tamper condition.
Magnet_Tamper	Magnetic tamper presence.
Communication	Communication activity with meter or module.
Relay_Connected	Load relay connected or closed.
Relay_Disconnected	Load relay disconnected or open.
Battery_Low	Low RTC or meter battery.
Reverse_Current	Reverse current or energy-flow direction.
Forward_Current	Forward current or energy-flow direction.
Neutral_Disturbance	Neutral disturbance condition.
Power_On	Supply power present.
Cover_Open	Top-cover or module-cover open tamper; presentation may be continuous or blinking.

- Descriptions are metadata for users and reviewers, not free-form prediction text.
- Blinking state must be handled only if the existing symbol subsystem supports temporal evidence.
- New symbols require catalog, schema and ontology version updates.

10 Unified JSON Schema

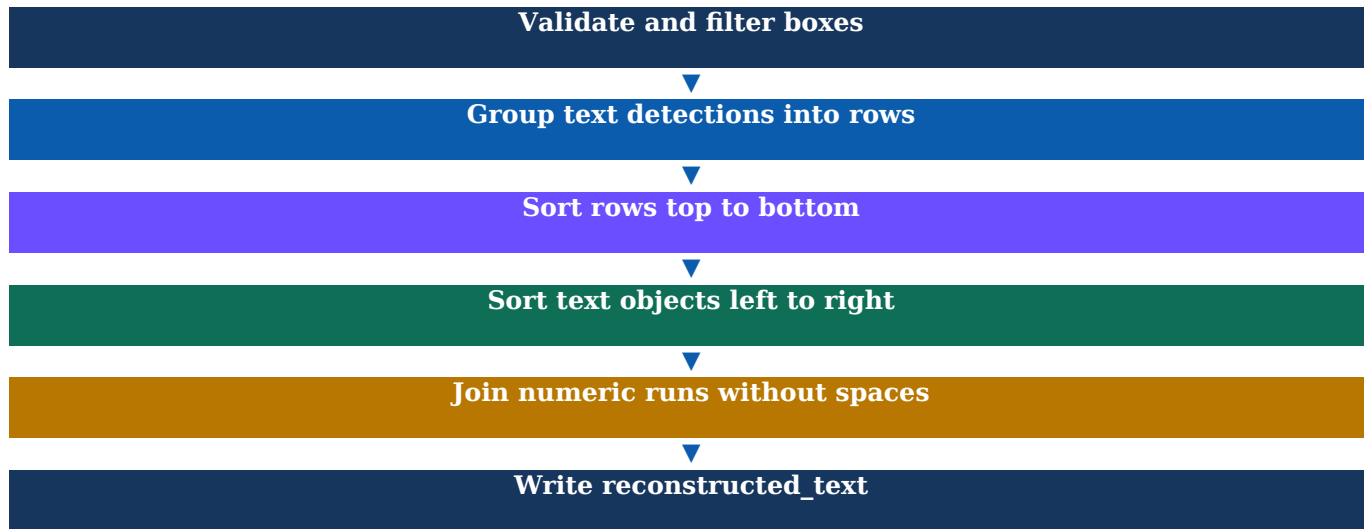
Common output for text and symbols

```
{
  "schema_version": "1.1",
  "image": "lcd_001.png",
  "width": 1280,
  "height": 720,
  "detections": [
    { "id": 1, "type": "text", "class": "PF", "bbox": [100,45,150,80]},
    { "id": 2, "type": "text", "class": "0", "bbox": [180,45,205,85]},
    { "id": 3, "type": "text", "class": ".", "bbox": [208,72,215,80]},
    { "id": 4, "type": "symbol", "class": "Battery_Low", "bbox": [420,25,460,58]}
  ],
  "reconstructed_text": "PF 0.",
  "symbols": ["Battery_Low"]
}
```

Field	Requirement
schema_version	Required string.
image, width, height	Required image metadata.
detections	Required array; may be empty only for verified blank display.
id	Unique integer inside image record.
type	text or symbol.
class	Approved ontology name.
bbox	[x1,y1,x2,y2] in pixels.
reconstructed_text	Derived from text detections only.
symbols	Canonical list derived from symbol detections.

11 Reading Order and Reconstruction

Ordering text without changing predictions

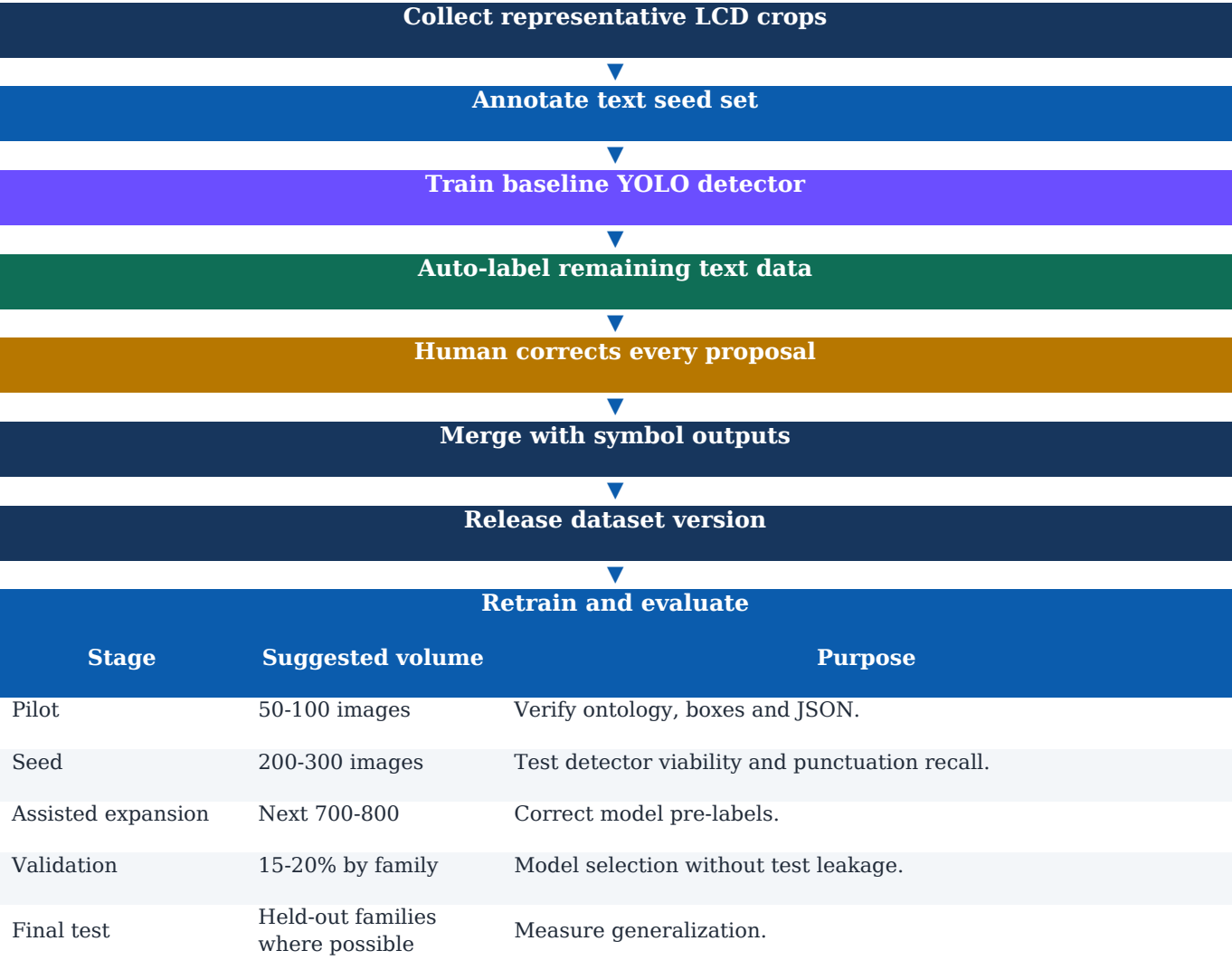


- Symbols remain structured objects and are not inserted into numeric strings.
- Fixed tokens are separated from numeric runs by one space.
- Multiple display rows use newline separators.
- No rule may invent missing decimal points, colons, digits or labels.
- Raw detector order is never trusted.

Row 1: [L1] [L2]
Row 2: [0] [2] [0] [6] [.] [3] [2]
Text: L1 L2\n0206.32
Symbols: [Battery_Low]

12 Dataset Lifecycle

Pilot, AI-assisted expansion and release



GO / NO-GO GATE

Do not annotate all 1,000 images before evaluating the 200-300 image seed model. Continue detector-only if punctuation recall and exact transcription show a credible path; otherwise test the hybrid fallback.

13 Evaluation Metrics and Quality Gates

Detector, symbol integration and end-to-end quality

Metric	Purpose	Priority
Per-class recall	Missed text or symbols.	Critical
Decimal recall	Missed decimal risk.	Critical
Colon recall	Missed time/date separator risk.	Critical
mAP@50-95	Box and class quality.	Secondary
Exact text accuracy	Whole reconstructed display text match.	Critical
Symbol exactness	Correct approved symbol set per image.	High
Unified output accuracy	Text and symbol content jointly correct.	Critical
Family-level accuracy	Generalization across meter families.	High
DATASET GATE		
100% of released JSON files must pass schema validation; boxes remain within image bounds; classes belong to the declared ontology.		
GENERALIZATION GATE		
Where available, the final report must include held-out display families. The final test set must not be repeatedly used for tuning.		

14 Versioning, Traceability and Risks

Reproducible lineage across both workstreams

Artifact	Example	Rule
Text ontology	text_ontology_v1.0	Immutable IDs.
Symbol catalog	symbol_catalog_v1.0	Canonical names and descriptions.
Unified schema	lcd_schema_v1.1	Version when fields or meaning change.
Dataset	lcd_dataset_v1.1	Images, text JSON, merged symbol results and splits.
Text model	yolo_text_v1.0	Records weights and training config.
Symbol model	symbol_model_existing	Referenced, not retrained by this scope.
Evaluation	eval_v1.1	Records test manifest and metrics.

Risk	Mitigation
Tiny punctuation	High input resolution, targeted examples, class recall reporting.
Class imbalance	Per-class counts and targeted collection.
Duplicate text/symbol boxes	Ownership rules and merge validation.
Coordinate mismatch	Require original-image pixel space for both subsystems.
Ontology drift	Version and lock IDs.
Data leakage	Split by family/session and deduplicate.

15 Implementation Plan and Roadmap

Final system baseline and controlled expansion

Step	Deliverable
1	Freeze text ontology and symbol catalog.
2	Validate unified JSON and coordinate conventions.
3	Annotate 50-100 image pilot.
4	Train YOLO small baseline.
5	Evaluate punctuation and exact text accuracy.
6	Auto-label and correct remaining data.
7	Merge existing symbol outputs.
8	Train candidate model and complete held-out evaluation.
9	Release model, schema, ontology, catalog and metrics.

Future phase	Addition
Phase 2	Parameter-value-unit semantic grouping.
Phase 3	Expansion to new symbol families.
Phase 4	Raw meter localization and rectification.
Phase 5	Blur, glare, perspective and illumination robustness.

FINAL REQUIREMENT

Given a cropped upright LCD image, the system shall return all visible approved text elements and approved symbols with class labels and bounding boxes, reconstruct the visible text without semantic interpretation, and generalize across LCD families without fixed coordinates or meter-specific templates.

- Approved text ontology and class mapping.
- Approved symbol catalog and descriptions.
- Validated unified JSON annotations.
- YOLO-compatible text export.
- Trained text detector.
- Integrated text-plus-symbol inference output.
- Evaluation report and reproducible version records.